

Does Fisher–Rao Birth–Death Help Adaptive Biasing Force?

A Controlled Two-Dimensional Study

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Abstract

We study whether augmenting the Adaptive Biasing Force (ABF) method with a Fisher–Rao (FR) birth–death correction improves the computation of a reaction-coordinate free energy, and, if so, how the correction must be constructed. On a controlled two-dimensional model potential with reaction coordinate $\xi(x, y) = x$ and a quadrature-exact reference free energy $F_{\text{ref}}(x)$, ABF estimates the mean force $F'(x) = \mathbb{E}[\partial_x V \mid X = x]$ while the FR step gently reallocates particles along x by killing replicas in over-represented regions and cloning replicas in under-represented regions, so that the empirical reaction-coordinate marginal $\hat{p}_t(x)$ is nudged toward a target marginal $q_t(x)$. We compare four methods — ABF only, and ABF plus FR with an *estimated*, *uniform*, or *oracle* target — across a production study of 545 GPU runs (109 configurations, 5 seeds, 1000 particles, 100,000 steps). The practical, estimated-target FR yields a *modest* improvement that is clearest in the time-integrated error $\int_0^T \|\hat{F}_t - F_{\text{ref}}\|_{L^2} dt$ (a reduction of 11.5% over ABF only at the best configuration), with smaller final-budget gains; FR is therefore better described as a convergence accelerator and stabiliser than as a rescue mechanism in this regime. The oracle target — which is allowed to use F_{ref} , the very quantity being computed — is reported *only as a diagnostic positive control*: it is not a usable algorithm, but the gap between estimated and oracle targets (11.5% versus 16.6% integrated-error reduction) indicates that practical performance is bounded by the quality and stability of the online free-energy target estimate. Across all successful settings the FR correction is weak (per-event particle-replacement fractions of order 10^{-5}), target-consistent, and does not visibly corrupt the conditional distribution $Y \mid X = x$ on which ABF depends.

1 Introduction

1.1 Free-energy computation along a reaction coordinate

A central task in computational statistical mechanics is to compute the free energy associated with a low-dimensional *reaction coordinate* of a high-dimensional system in thermal equilibrium [6]. Let $q = (x, y) \in \mathbb{R}^2$ denote a configuration¹ drawn from the canonical (Boltzmann–Gibbs) density

$$\pi(x, y) \propto e^{-\beta V(x, y)}, \quad (1)$$

where V is the potential energy and $\beta = 1/(k_B T)$ is the inverse temperature. Fixing the reaction coordinate to the first coordinate, $\xi(x, y) = x$, the free energy $F(x)$ and the (local) mean force $F'(x)$

¹We write a configuration as $q = (x, y)$. The symbol $q_t(x)$, always carrying a time subscript and a scalar argument, denotes instead the Fisher–Rao *target marginal* of Section 3; the two are never used in the same role.

along ξ are the objects of interest. The difficulty is metastability: when V has well-separated basins along x , unbiased Langevin dynamics crosses the barrier rarely, so a direct estimate of F converges intolerably slowly.

1.2 Adaptive Biasing Force

The Adaptive Biasing Force (ABF) method [2–4] removes this metastability adaptively. It maintains a running estimate $\widehat{F}'_t(x)$ of the mean force and applies the antiderivative $B_t(x) = \int^x \widehat{F}'_t$ as a bias that depends only on x . As $B_t \rightarrow F$, the effective dynamics along x becomes diffusive and the reaction coordinate is explored freely, while — crucially — the bias does not change the conditional law of the orthogonal degrees of freedom given x . ABF is mathematically well understood, with long-time convergence guarantees [5], and is a natural baseline against which to measure any proposed enhancement.

1.3 Why add a Fisher–Rao correction?

ABF accelerates exploration through the *force*, but it does not directly control *where the particles are*. Early in a simulation the estimated bias is poor, so the empirical reaction-coordinate marginal $\widehat{p}_t(x)$ can remain concentrated in the initial basin, starving other values of x of the samples ABF needs to estimate $F'(x)$ there. A Fisher–Rao (FR) birth–death correction [7] addresses exactly this: interpreting the evolution of $\widehat{p}_t$ as a gradient flow of relative entropy in the Fisher–Rao geometry [1], one reallocates probability mass by *killing* particles in over-represented regions and *cloning* particles in under-represented regions, driving $\widehat{p}_t$ toward a chosen target marginal $q_t(x)$ without integrating any additional dynamics. Coupling such a correction to ABF is appealing because the two act on complementary objects (force versus density), but it is not obvious that it helps: a birth–death step resamples *whole* configurations (x, y) , and if it distorts the conditional law $Y \mid X = x$ it will corrupt precisely the quantity ABF estimates.

1.4 Central question and contributions

This report asks three questions on a controlled example where the exact answer F_{ref} is available by quadrature:

1. **Does FR help ABF?** (Q1)
2. **How should FR be added so that it helps?** (Q2–Q3)
3. **What are the limitations, failure modes, and future directions?** (Q4)

Our contributions are: (i) a precise statement of the coupled ABF–FR algorithm in consistent notation (Sections 2 and 3); (ii) a large production study (545 runs) that quantifies, with matched seeds, the effect of FR and of its target marginal (Sections 4 and 5); and (iii) a mechanistic analysis (Section 6) separating the contribution of the FR *mechanism* from that of the FR *target*. A recurring methodological point is the careful interpretation of an *oracle* target that uses F_{ref} : it is a diagnostic control that bounds the achievable benefit, never a deployable method. We find that the practical, estimated-target FR is a *modest convergence accelerator* here, and we delineate the conditions under which it helps and those under which it could hurt.

2 Background: ABF for a reaction-coordinate free energy

2.1 Free energy and the unbiased marginal

With reaction coordinate $\xi(x, y) = x$, the unbiased reaction-coordinate marginal of the canonical density (1) is

$$p_{\text{ref}}(x) = \frac{\int e^{-\beta V(x, y)} dy}{\iint e^{-\beta V(x, y)} dx dy}, \quad (2)$$

and the free energy along ξ is, up to an additive constant,

$$F_{\text{ref}}(x) = -\frac{1}{\beta} \log p_{\text{ref}}(x) + C. \quad (3)$$

Because F is defined only up to the constant C , every free-energy profile in this report — both estimates and the reference — is *centred* (its mean over the interior evaluation window is subtracted) before any L^2 error is formed; otherwise a meaningless constant offset would dominate the comparison. The reference quantities are computed by a one-dimensional y -quadrature on a fine grid with a log-sum-exp shift for numerical stability, which is exact up to discretisation and provides the ground truth $F_{\text{ref}}, F'_{\text{ref}}, p_{\text{ref}}$.

2.2 The mean force is the object ABF estimates

Differentiating (3) and using $\partial_x p_{\text{ref}}$ gives the *mean force*

$$F'_{\text{ref}}(x) = \frac{\int \partial_x V(x, y) e^{-\beta V(x, y)} dy}{\int e^{-\beta V(x, y)} dy} = \mathbb{E}_{\pi}[\partial_x V(X, Y) | X = x]. \quad (4)$$

Equation (4) is the cornerstone of ABF: the gradient of the free energy equals the *conditional expectation of the instantaneous force* $\partial_x V$ given the reaction coordinate. It is a local, sample-based quantity — no integral of V is required — so it can be estimated on the fly from the particles currently at $X \approx x$, and the free energy is then recovered by quadrature, $\hat{F}(x) = \int^x \hat{F}'$.

2.3 Biased dynamics

ABF simulates N overdamped Langevin replicas that share one running mean-force estimate $\hat{F}'_t(x) \approx F'_{\text{ref}}(x)$ with bias potential $B_t(x) = \int^x \hat{F}'_t$. Each replica (X_t, Y_t) evolves as

$$\begin{aligned} dX_t &= [-\partial_x V(X_t, Y_t) + \hat{F}'_t(X_t)] dt + \sqrt{2/\beta} dW_t^x, \\ dY_t &= -\partial_y V(X_t, Y_t) dt + \sqrt{2/\beta} dW_t^y, \end{aligned} \quad (5)$$

with independent Brownian motions W^x, W^y . The added force $+\hat{F}'_t(X_t)$ acts *only* in the x -direction; the y -dynamics is unbiased. When $\hat{F}'_t \rightarrow F'_{\text{ref}}$ the net drift along x becomes $-\partial_x(V - F)$, whose x -marginal equilibrium is flat, so the reaction coordinate diffuses freely across the barrier.

2.4 Why the bias does not spoil the mean-force estimate

A bias that depends only on x leaves the conditional law of Y given $X = x$ invariant. This is what makes ABF self-consistent.

Proposition 1 (Conditional invariance under an x -only bias). *Let ν be the canonical measure $\nu \propto e^{-\beta V(x,y)}$ and let $\nu_B \propto e^{-\beta [V(x,y)-B(x)]}$ be the measure biased by any function B of x alone. Then for every x the conditionals coincide, $\nu_B(y \mid X = x) = \nu(y \mid X = x)$, and consequently*

$$\mathbb{E}_{\nu_B}[\partial_x V(X, Y) \mid X = x] = \mathbb{E}_{\nu}[\partial_x V(X, Y) \mid X = x] = F'_{\text{ref}}(x). \quad (6)$$

Proof. At fixed $X = x$ the bias factor $e^{\beta B(x)}$ is a constant in y and cancels between numerator and normaliser of the conditional density:

$$\nu_B(y \mid X = x) = \frac{e^{-\beta [V(x,y)-B(x)]}}{\int e^{-\beta [V(x,y')-B(x)]} dy'} = \frac{e^{-\beta V(x,y)}}{\int e^{-\beta V(x,y')} dy'} = \nu(y \mid X = x).$$

Taking the conditional expectation of $\partial_x V$ under this common conditional gives (6), the last equality being (4). $\square$

Proposition 1 is the reason ABF works: even though the biased ensemble (5) has a different x -marginal than the canonical one, the conditional samples $Y \mid X = x$ remain canonical, so the empirical conditional average of $\partial_x V$ is an unbiased target for $F'_{\text{ref}}(x)$. Any modification of ABF must respect this property; in particular, a birth–death correction that resamples whole configurations (x, y) is only admissible if it too leaves $Y \mid X = x$ (approximately) intact — a hypothesis we test directly in Sections 5 and 6.

2.5 The ABF mean-force estimator

In the kernel form used by the reference implementation, $\hat{F}'$ is a Nadaraya–Watson running average of the instantaneous force,

$$\hat{F}'_n(x) = \frac{\sum_{k \leq n} \sum_i K_h(x - X_k^i) \partial_x V(X_k^i, Y_k^i)}{\sum_{k \leq n} \sum_i K_h(x - X_k^i)}, \quad (7)$$

with a Gaussian kernel K_h of bandwidth h , accumulated over all replicas i and past steps k ; the free-energy estimate is its pinned trapezoidal antiderivative, $\hat{F}_n(x) = \int_0^x \hat{F}'_n$, and we use $B_t \equiv \hat{F}$ as the bias. The GPU backend used for the production study replaces the $O(n_{\text{grid}} N)$ kernel sums by an $O(N + n_{\text{grid}})$ *binned-smoothed* estimator: particles are scattered onto the x -grid and the count and force histograms are Gaussian-smoothed, so that $\hat{F}'(x) = \text{smooth}(\text{force})/\text{smooth}(\text{count})$ is a Riemann-sum approximation of (7) that converges to it as the grid spacing vanishes. The two estimators are validated to agree to well under 1% on static tests (Section 4).

3 The Fisher–Rao marginal correction

The Fisher–Rao (FR) correction is a birth–death / particle-reallocation step that acts on the *reaction-coordinate marginal* of the ensemble. It is interleaved with the ABF dynamics (5) and is designed to drive the empirical marginal $\hat{p}_t(x)$ toward a target marginal $q_t(x)$, without adding any drift to the (x, y) dynamics.

3.1 Empirical and target marginals

At step n , after the Langevin proposal of all N replicas, the empirical reaction-coordinate marginal is estimated by a (reflected, edge-corrected) kernel density / binned-smoothed estimate of bandwidth η ,

$$\hat{p}_n(x) = \frac{1}{N} \sum_{i=1}^N K_\eta(x - X_n^i), \quad (8)$$

renormalised on the x -grid. The target $q_n(x)$ is one of three choices (Section 3.5); all are nonnegative and normalised, $\int q_n = 1$.

3.2 Fisher–Rao score

For each particle the FR *score* is the pointwise log-ratio of the empirical to the target marginal, with its ensemble mean removed:

$$S_i = \log \frac{\hat{p}_n(X_n^i) + \varepsilon}{q_n(X_n^i) + \varepsilon} - \underbrace{\int \hat{p}_n(z) \log \frac{\hat{p}_n(z)}{q_n(z)} dz}_{= \text{KL}(\hat{p}_n \| q_n)}, \quad (9)$$

where $\varepsilon > 0$ is a small regulariser and the integral baseline equals the Kullback–Leibler divergence $\text{KL}(\hat{p}_n \| q_n)$ (a grid quadrature; it coincides with the Monte-Carlo baseline $\frac{1}{N} \sum_k \log[\hat{p}_n(X^k)/q_n(X^k)]$ up to discretisation). Subtracting the baseline makes the score mean-zero, so the birth–death step below conserves the expected particle number. The score is clipped to $[-S_{\max}, S_{\max}]$ for stability. Its interpretation is direct:

- $S_i > 0$: X^i lies where $\hat{p}_n > q_n$, an *over-represented* region (too many particles);
- $S_i < 0$: X^i lies where $\hat{p}_n < q_n$, an *under-represented* region (too few particles).

3.3 Birth–death resampling

The FR step realises the Fisher–Rao gradient flow of $\text{KL}(\hat{p} \| q)$ as a fixed- N birth–death process. Over a sub-step of length Δt and at effective rate γ_n^{eff} (Section 3.4), each *over-represented* particle ($S_i > 0$) is killed with probability

$$\mathbb{P}(i \text{ dies}) = 1 - \exp(-\gamma_n^{\text{eff}} S_i \Delta t), \quad (10)$$

and every killed slot is overwritten by a *clone* of an *under-represented* particle, sampled from the pool $\{i : S_i < 0\}$ with weights $\propto |S_i|$. Cloning copies the *whole* configuration (X^i, Y^i) — the score depends only on x , but the y -coordinate must travel with its x -coordinate, or Proposition 1 would be violated. The step is made deliberately gentle by three safeguards: (i) a burn-in and soft ramp on the rate; (ii) score clipping; and (iii) a hard cap $f_{\max}$ on the fraction of particles replaced per event (`max_event_fraction`), enforced by randomly sub-sampling deaths when the cap is exceeded. Net effect: gently replace particles in crowded regions of x by copies from starved regions, leaving N fixed.

3.4 Rate schedule

The FR rate is zero during an initial burn-in of $n_{\text{burnin}} = \lfloor b n_{\text{steps}} \rfloor$ steps (burn-in fraction b), then ramps softly to its maximum γ ,

$$\gamma_n^{\text{eff}} = \gamma \left(1 - e^{-(n - n_{\text{burnin}})/n_{\text{ramp}}} \right), \quad n > n_{\text{burnin}}, \quad (11)$$

and the birth-death step is applied every `fr_every` steps. The burn-in lets ABF build a usable mean-force estimate (hence a usable target) before any particles are moved; the ramp avoids a sudden shock to the ensemble.

3.5 Target marginals: estimated, uniform, oracle

The three targets differ only in which density q_n the score (9) is taken against. Recall $B_t = \hat{F}_n$ is the current ABF bias.

Estimated target (the practical method). A target free energy $\hat{F}_n^{\text{target}}$ is maintained independently of the bias by an exponential moving average of the ABF estimate, $\hat{F}_{n+1}^{\text{target}} = (1 - r) \hat{F}_n^{\text{target}} + r \hat{F}_{n+1}$, and the target marginal is the *biased* marginal that this target free energy would induce under the current bias,

$$q_n^{\text{est}}(x) \propto \exp \left[-\beta (\hat{F}_n^{\text{target}}(x) - B_t(x)) \right]. \quad (12)$$

This uses only quantities available at run time and is the method whose practical value we assess.

Uniform target (a naive ablation).

$$q_n^{\text{unif}}(x) = \text{Unif}(x) \quad \text{on the } x\text{-grid}. \quad (13)$$

This tests whether simply *flattening* the reaction-coordinate marginal is enough. It is a target ablation, not the intended method; it ignores the fact that, under a bias $B_t \neq F$, the canonical biased marginal is not uniform.

Oracle target (a diagnostic control only).

$$q_n^{\text{oracle}}(x) \propto \exp \left[-\beta (F_{\text{ref}}(x) - B_t(x)) \right]. \quad (14)$$

Remark 1 (The oracle target is not an algorithm). Equation (14) replaces the estimated target free energy by the *reference* free energy F_{ref} — which is exactly the unknown quantity the whole procedure is trying to compute. The oracle target therefore *cannot* be evaluated in any real free-energy calculation. We include it solely as a *diagnostic / positive / ideal-information control*: it measures how much the FR mechanism *could* help if the target marginal direction were specified perfectly. Throughout this report the oracle is read only in that way; we never claim that “oracle FR is the best method” or that “one should use the oracle target”. The practical comparison is always

$$\text{ABF only} \quad \text{versus} \quad \text{ABF} + \text{estimated-target FR},$$

and the estimated-versus-oracle gap is interpreted as the headroom available from improving the online target estimate.

3.6 The coupled ABF–FR step

One outer iteration, given the state $\{(X_n^i, Y_n^i)\}$ and accumulators, is:

1. update the ABF accumulators from the current particles and recompute $\widehat{F}'_n, \widehat{F}_n = B_t$ (Section 2.5);
2. propose the biased Langevin move (5) for all replicas;
3. if FR is active at this step (past burn-in, on the `fr_every` schedule): form $\widehat{p}_n$ (8) and the target q_n — one of (12), (13), (14) — compute the scores (9), and apply the birth–death step (10);
4. record diagnostics at the evaluation cadence.

The ABF estimator sees the post-resampling ensemble at the next step, which is the route by which improved x -coverage feeds back into a better mean-force estimate. The remainder of the report quantifies whether, and how, this feedback is beneficial.

4 Experimental design

4.1 Model potential and reference

We use a two-dimensional model potential with reaction coordinate $\xi(x, y) = x$,

$$V(x, y) = 3e^{-x^2} \left(e^{-(y-1/3)^2} - e^{-(y-5/3)^2} \right) - 5e^{-y^2} \left(e^{-(x-1)^2} + e^{-(x+1)^2} \right) + 0.2x^4 + 0.2(y - 1/3)^4. \quad (15)$$

As shown in Figure 1, V has two metastable basins near $x = \pm 1$ separated by a barrier near $x = 0$ (with an upper sub-basin near $(0, 5/3)$), so unbiased dynamics is metastable in x . We fix $\beta = 4$. The reference $F_{\text{ref}}, F'_{\text{ref}}, p_{\text{ref}}$ are obtained by the y -quadrature of (2)–(4) on a fine grid and are treated as ground truth.

4.2 Methods compared

Four methods are compared, identical except for the FR target: **ABF only** (no birth–death; baseline); **ABF+FR estimated** (12) (the practical method); **ABF+FR uniform** (13) (naive marginal-flattening ablation); and **ABF+FR oracle** (14) (diagnostic control, Remark 1). All four share the same potential, integrator, ABF estimator, seeds and initial conditions, so differences are attributable to the FR step and its target alone (matched-seed comparison).

4.3 Metrics

All profile errors are interior, domain-RMS L^2 distances on the window $x \in [-2.5, 2.5]$ (the reflecting walls and the x^4 confinement make the edges uninformative), with free-energy profiles centred as in Section 2.

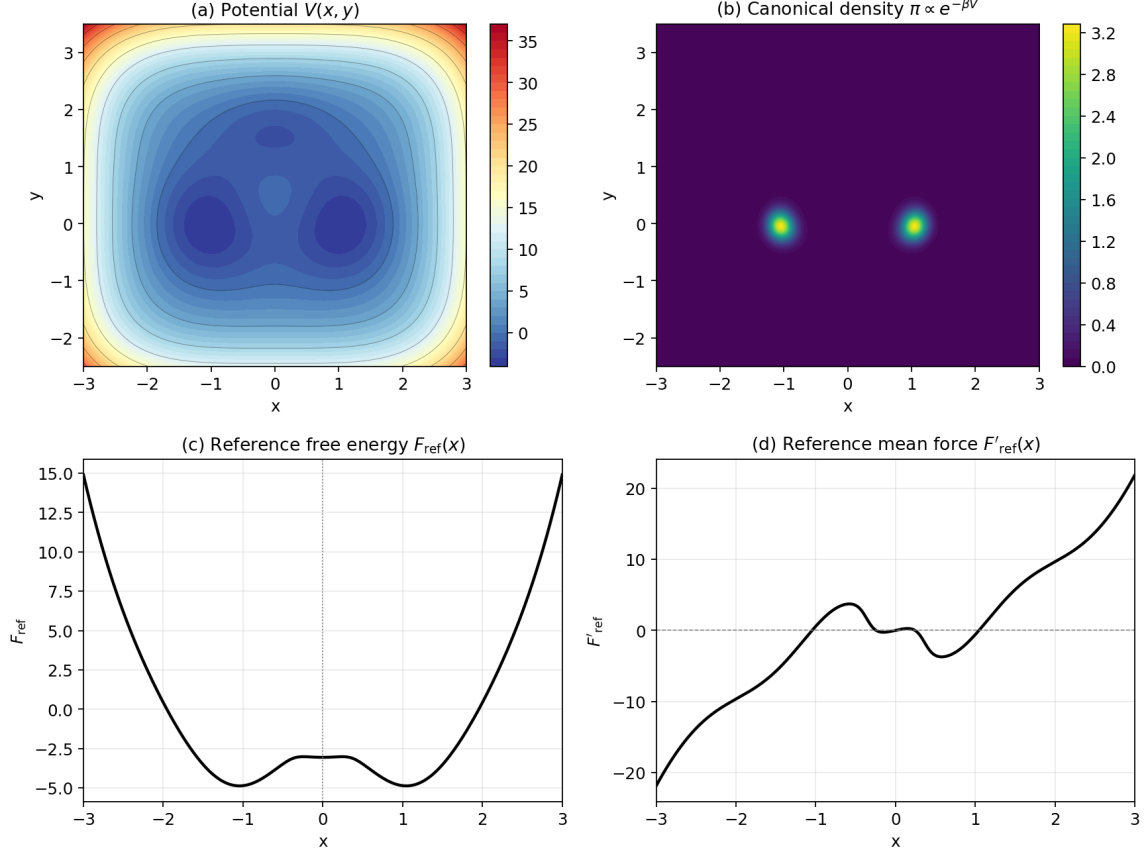


Figure 1: Problem geometry. (a) The potential $V(x, y)$ (15) with two basins near $x = \pm 1$. (b) The canonical density $\pi \propto e^{-\beta V}$ at $\beta = 4$, concentrated in the two basins. (c) The reference free energy $F_{\text{ref}}(x)$ (a double well in x) and (d) the reference mean force $F'_{\text{ref}}(x)$, both by y -quadrature.

Primary (practical) metrics.

$$\left\| \hat{F}_T - F_{\text{ref}} \right\|_{L^2}, \quad \left\| \hat{F}'_T - F'_{\text{ref}} \right\|_{L^2}, \quad \int_0^T \left\| \hat{F}_t - F_{\text{ref}} \right\|_{L^2} dt. \quad (16)$$

The first two are *final-budget accuracy* at the end of the run; the *integrated* error rewards *fast* convergence (anytime performance), since a method that reaches a good profile earlier accumulates less error over $[0, T]$. These two notions need not be optimised by the same hyperparameters, a point we treat carefully in Section 5.

Diagnostic metrics. Reaction-coordinate marginal flatness ($\|\hat{p} - \text{Unif}\|$ and $\|\hat{p} - q\|$); cumulative barrier crossings at $x = 0$; region fractions (left basin / barrier / right basin); the FR event fraction (particles replaced per FR application) and the FR score dispersion $\text{std}_i S_i$; conditional $Y \mid X = x$ errors (Section 4.3); and ancestor/diversity counts. We stress that *marginal flatness is a diagnostic, not the objective*: the goal is free-energy/mean-force accuracy (16). A method can flatten $\hat{p}$ and yet fail to improve $\hat{F}$ if it does so by corrupting the conditional samples.

Conditional $Y \mid X$ diagnostics. At a set of x -bin centres we compare the empirical conditional $\hat{p}(y \mid x \in I_j)$ (the y -histogram of particles whose x falls in a narrow window) to the reference

Table 1: Production study design (read from the result CSVs and configs/two_dim_xi_x_production_gpu.yaml).

Quantity	Value
Inverse temperature β	4.0
Time step Δt	0.002
Steps per run	100,000
Particles per run N	1000
Seeds	5 (0, 1, 2, 3, 4)
Methods	4 (ABF; FR estimated / uniform / oracle)
FR rate γ	0.01, 0.02, 0.05, 0.1
FR bandwidth η	0.075, 0.1, 0.15
FR burn-in fraction	0, 0.2, 0.4
FR stride <code>fr_every</code>	5
Total configurations	109
Total runs	545

Table 2: Hyperparameters of the selected configurations (integrated- $L^2(F)$ rule). n_{FR} is the Fisher–Rao stride `fr_every`.

Method	target	γ	η	burn-in	n_{FR}	seeds
ABF only	none	0	0.075	0	1	5
FR estimated	estimated	0.1	0.075	0	5	5
FR uniform	uniform	0.05	0.15	0.2	5	5
FR oracle*	oracle	0.1	0.075	0	5	5

conditional $p_{\text{ref}}(y \mid x) \propto e^{-\beta V(x,y)}$, via an L^2 distance and a KL divergence. By Proposition 1 a benign FR step should leave these close to the ABF-only values.

4.4 GPU implementation and validation

The production runs use a batched PyTorch backend that simulates many runs at once and reuses the reference metrics/diagnostics code verbatim. For speed it uses the binned-smoothed ABF/FR estimator of Section 2.5 rather than the exact kernel sums (7), and a fully vectorised fixed- N birth–death. The two backends are not bit-identical (RNG and floating-point order differ), but a validation suite passes all of its checks: the analytic torch gradient matches finite differences and the numpy gradient; numpy-CPU versus torch-CPU and torch-CPU versus torch-CUDA metrics agree to the same order of magnitude; batch-size invariance holds; the binned-smooth estimator matches the kernel reference to within the stated tolerance ($< 10\%$ static, well under 1% in practice); and the FR event-fraction cap is respected. We flag in Section 7 that this binned estimator is a deliberate, but not mathematically identical, surrogate for the kernel notebook implementation.

4.5 Hyperparameter grid and study size

The FR methods sweep the rate γ , the marginal bandwidth η , and the burn-in fraction b , at a fixed FR stride; ABF only has no FR parameters. The resulting study (Table 1) comprises 545 runs over 109 configurations and 5 seeds, at 1000 particles and 100,000 steps each. We verified directly from the result CSVs that all 545 runs completed with *no* NaNs in the primary metrics. The selected configurations reported in Section 5 are listed in Table 2.

Table 3: Main comparison, configurations selected by smallest median *integrated* $L^2(F)$ (convergence speed). Lower is better except P (beats ABF).

Method	$\int_0^T \ \hat{F}_t - F_{\text{ref}}\ dt$	$\ \hat{F}_T - F_{\text{ref}}\ $	$\ \hat{F}'_T - F'_{\text{ref}}\ $	$P(\text{beats ABF})$	FR event frac.
ABF only	15.91	0.0137	0.0718	—	0
FR estimated	14.08	0.0126	0.0771	0.60	1.6×10^{-5}
FR uniform	14.46	0.0125	0.0769	0.80	5.3×10^{-6}
FR oracle*	13.27	0.0113	0.0602	1.00	1.7×10^{-5}

* The oracle target uses the reference free energy F_{ref} (the unknown being computed); it is a diagnostic positive control, *not* a usable algorithm. Per-config medians over 5 seeds; one configuration per method, selected by the smallest median integrated $L^2(F)$.

5 Results: does Fisher–Rao help ABF?

5.1 Headline comparison

Table 3 reports the four methods, each at its best configuration under the *integrated*- $L^2(F)$ rule (convergence speed; see Section 5.4). Reading the practical comparison (ABF only versus ABF+FR estimated): the estimated-target FR reduces the median integrated free-energy error from 15.91 to 14.08, a 11.5% reduction, while the *final* free-energy error improves only slightly (from 0.0137 to 0.0126). The birth–death is extremely gentle: the median per-event replacement fraction is 1.6×10^{-5} , i.e. of order one particle in 10^5 per FR application. Figure 2 shows the same story over time: the FR curves (estimated, uniform, oracle) detach below the ABF-only curve in the early and middle phase of the $L^2(F)$ panel and the ensemble re-equilibrates faster, with the bands (inter-quartile range over seeds) overlapping by the end of the budget.

5.2 Final profiles

Figure 3 overlays the final mean-force, free-energy and reaction-coordinate-marginal profiles. All practical methods recover F'_{ref} and F_{ref} well; the visible differences between ABF only and ABF+FR estimated at the final budget are small, consistent with the modest final-error improvements in Table 3. The x -marginal (panel c) is somewhat flatter under FR, as intended, but — as we emphasise in Section 4.3 — marginal flatness is a diagnostic, not the goal.

5.3 Target-type ablation

Figure 4 shows the distribution, over the 36 configurations of each target type, of the seed-median errors. Three honest observations follow. First, for the *integrated* error all three FR target types have medians below the ABF-only baseline (dashed line) — FR typically accelerates convergence — but the spread is wide and some configurations are *worse* than ABF, so FR does not help unconditionally. Second, for the *final* $L^2(F)$ the estimated and uniform medians sit essentially on the ABF baseline: the practical final-budget gain is marginal. Third, and most telling, on the final *mean-force* error the estimated and uniform medians lie slightly *above* the ABF baseline (a small degradation at the typical configuration), whereas only the oracle control is consistently below it. In other words, flattening the marginal with an imperfect target buys early speed but not final mean-force accuracy; only an accurate target direction (the oracle) improves everything at once.

Convergence of the selected (best integrated $L^2(F)$) configuration: median and IQR over 5 seeds

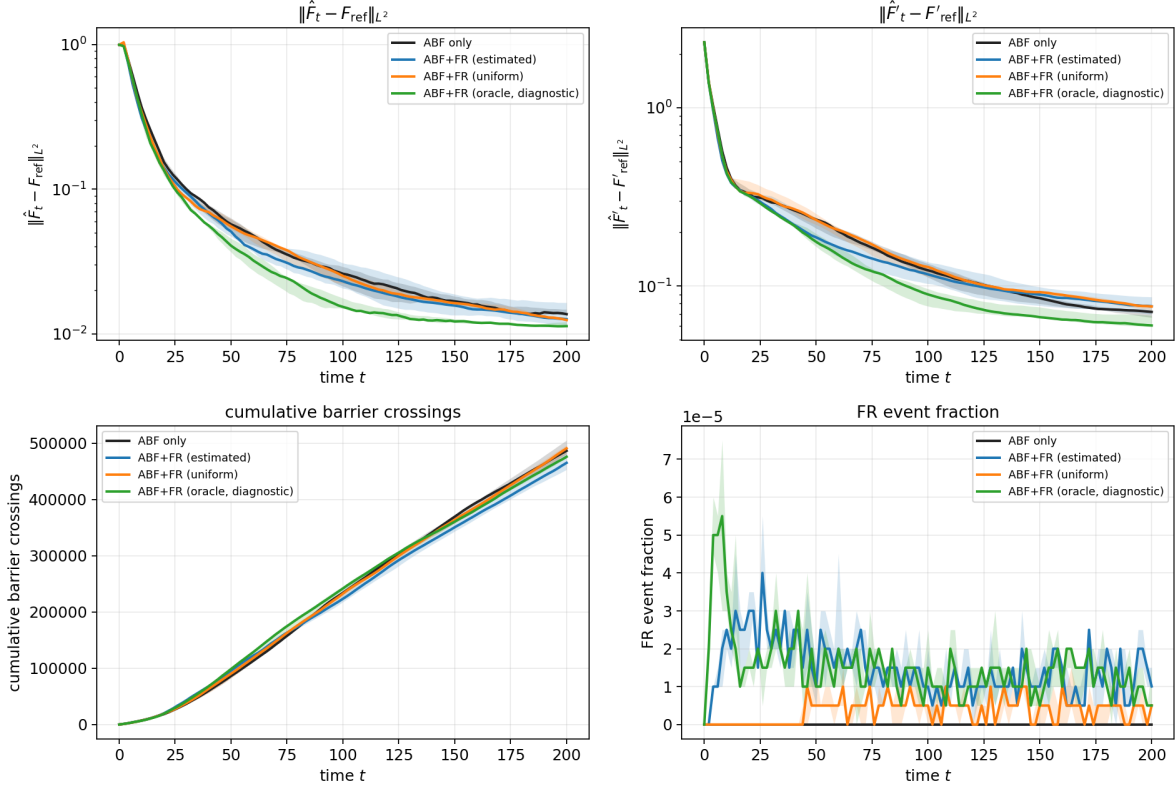


Figure 2: Convergence of the selected (integrated- $L^2(F)$ rule) configuration for each method: median (line) and inter-quartile range (band) over 5 seeds. The free-energy and mean-force errors (top, log scale) fall fastest for the oracle diagnostic and faster for the estimated/uniform FR than for ABF only in the early-to-middle phase; barrier crossings (bottom left) are essentially indistinguishable, and the FR event fraction (bottom right) is of order 10^{-5} throughout — FR is acting as a weak correction, not a violent resampling. The oracle curve is a diagnostic control (Remark 1).

5.4 Two selection rules give different “best” hyperparameters

Because integrated and final errors measure different things, they need not be optimised by the same configuration, and conflating them would be misleading. We therefore report two rules explicitly:

- selecting by *integrated* $L^2(F)$ (Table 3, the main text) favours *fast convergence*; for the estimated target it selects an *aggressive* configuration ($\gamma = 0.1$, burn-in 0);
- selecting by *final* $L^2(F)$ (Table 4, supplement) favours *final-budget accuracy*; for the estimated target it selects a *gentle, delayed* configuration ($\gamma = 0.01$, burn-in 0.4), which improves the final free-energy error by 12.5% and the final mean-force error by 6.3% over ABF only, and beats ABF on 80.0% of matched seeds.

This dependence on the selection rule is itself a finding: an aggressive FR rate accelerates the integrated error but does not (here) improve final accuracy, while a gentle, late FR is the safer choice for final accuracy. We emphasise the integrated rule in the main text because the clearest, most robust effect of FR in this example is on convergence *speed*; we are explicit that the final-budget improvements are modest.

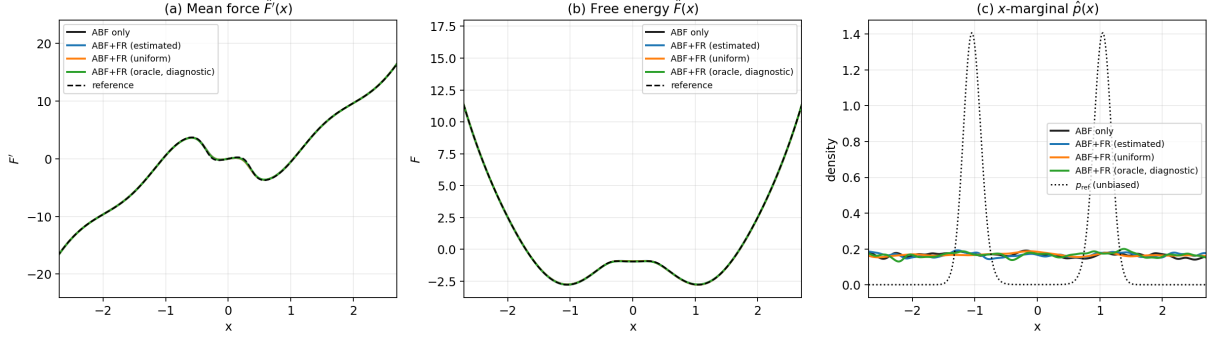


Figure 3: Final profiles (mean over seeds) for the selected configurations. (a) Mean force $\hat{F}'(x)$ and (b) centred free energy $\hat{F}(x)$ versus the reference (dashed); (c) reaction-coordinate marginal $\hat{p}(x)$, with the unbiased p_{ref} (dotted) for reference. Practical methods recover the reference well; FR mildly flattens $\hat{p}$.

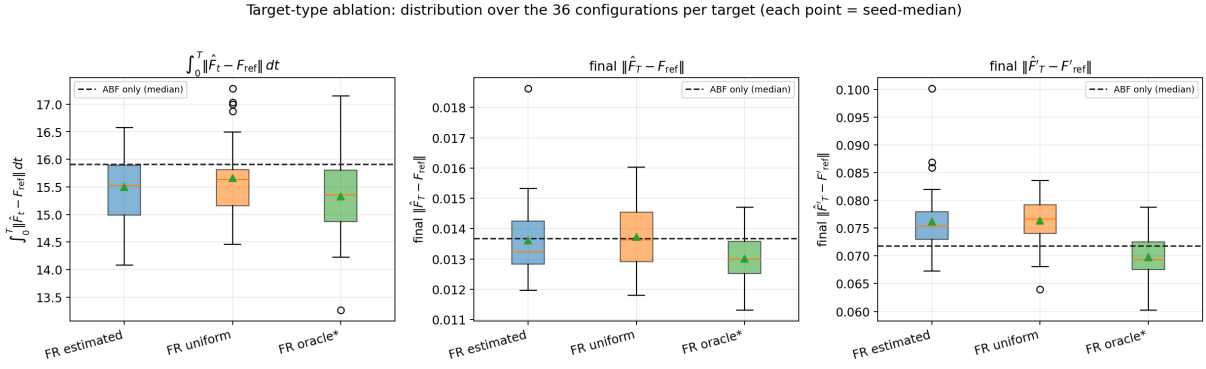


Figure 4: Target-type ablation: each box is the distribution over the 36 configurations of a target type (each point a seed-median); the dashed line is the ABF-only median. FR lowers the *integrated* error (left) but the practical targets barely change the *final* $L^2(F)$ (middle) and slightly raise the final mean-force error (right) at the median configuration. Only the oracle diagnostic improves all three.

5.5 The uniform target is a competitive naive ablation

In this comparatively easy example the *uniform* target is competitive with the estimated target (integrated reduction 9.1%; best final $L^2(F) = 0.0118$). We do not read this as evidence that uniform flattening is a general solution: it is a naive ablation that happens to work here because, under a near-converged bias $B_t \approx F$, the canonical biased marginal is already close to uniform, so “flatten $\hat{p}$ ” and “match the biased marginal” nearly coincide. In harder regimes (Section 7) the two targets would diverge and the uniform target would be expected to misdirect the birth–death.

5.6 The oracle control indicates headroom, not a method

The oracle diagnostic (Remark 1) achieves the lowest errors of all — integrated reduction 16.6%, final reduction 17.3%, and it beats ABF on 100.0% of matched seeds. *This is not a claim that one should use the oracle target*; the oracle is unusable because it requires F_{ref} . The correct reading is comparative: the estimated target captures 11.5% of the integrated-error reduction while the oracle reaches 16.6%, so a substantial part of the potential benefit of the FR mechanism is left unrealised by

Table 4: Supplement: configurations selected by smallest median *final* $L^2(F)$ (final-budget accuracy).

Method	$\int_0^T \ \hat{F}_t - F_{\text{ref}}\ dt$	$\ \hat{F}_T - F_{\text{ref}}\ $	$\ \hat{F}'_T - F'_{\text{ref}}\ $	$P(\text{beats ABF})$	FR event frac.
ABF only	15.91	0.0137	0.0718	—	0
FR estimated	15.25	0.0120	0.0673	0.80	1.1×10^{-6}
FR uniform	14.97	0.0118	0.0680	0.60	7.3×10^{-6}
FR oracle*	13.27	0.0113	0.0602	1.00	1.7×10^{-5}

* Oracle is a diagnostic control (uses F_{ref}), not a deployable method. Configurations here are selected by smallest median *final* $L^2(F)$; cf. Table 3, which need not pick the same hyperparameters.

the current online target estimate. The oracle–estimated gap thus localises the practical bottleneck in the *target estimation*, not in the birth–death mechanism itself.

5.7 Answer to Q1

On this example: *yes, but modestly, and chiefly as a convergence accelerator*. Practical estimated-target FR improves the time-integrated free-energy error appreciably (11.5%) and gives a smaller final-budget improvement; the gain is real but not a dramatic rescue, and it is not unconditional — poorly chosen FR configurations match or underperform ABF. FR is therefore best described, in this regime, as a *convergence accelerator and stabiliser* rather than a method that qualitatively changes what ABF can compute. The oracle control shows that more is possible *if* the target can be estimated better.

6 Mechanistic analysis

6.1 How does FR help? (Q2)

The mechanism is a feedback loop between coverage and the mean-force estimate.

1. ABF needs samples *across* the reaction coordinate: by (4), $F'_{\text{ref}}(x)$ is a conditional average that can only be estimated where particles actually visit x .
2. Early in a run the bias is poor and the ensemble is unbalanced — Figure 5(b) shows the initial occupancy split (~ 0.5 in one basin) relaxing only gradually toward the balanced 1/3–1/3–1/3 split across left basin, barrier and right basin.
3. FR compares the empirical marginal $\hat{p}_t$ with the target q_t : particles in over-represented regions of x receive positive score (9) and are more likely to be killed, while those in under-represented regions receive negative score and are more likely to be cloned (10).
4. This raises the useful coverage / sampling balance of x earlier than diffusion alone would. Figure 5(a) shows the estimated-FR free-energy error sitting below ABF only in the early window, and the FR score dispersion $\text{std}_i S_i$ in panel (d) decaying as $\hat{p}_t$ is pulled toward q_t .
5. Better coverage feeds back into a better conditional mean-force estimate, *provided* the birth–death does not corrupt $Y | X = x$. By Proposition 1 a benign FR step preserves the conditional, so improved x -coverage translates into a better $\hat{F}'$ and hence a better $\hat{F}$.

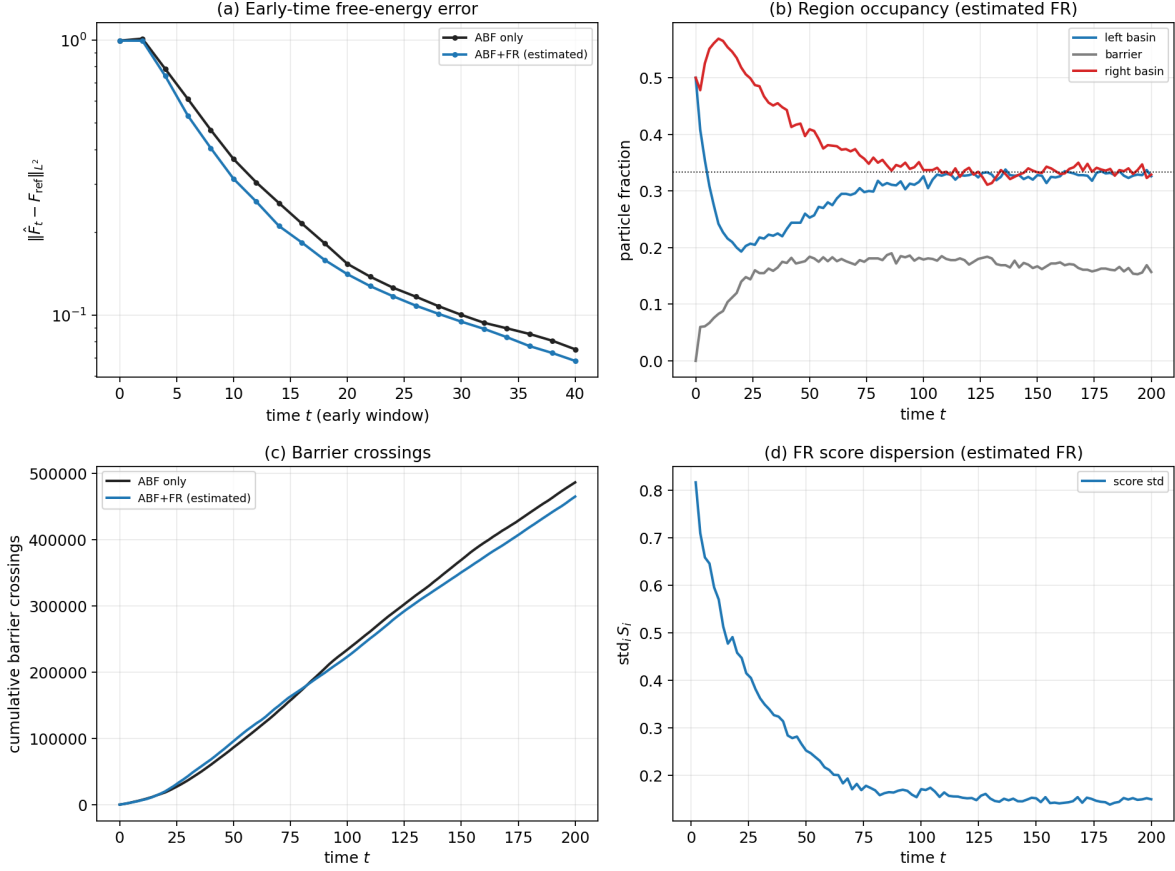


Figure 5: Mechanistic / early-time behaviour, estimated FR (best integrated-rule config) versus ABF only. (a) Early-time free-energy error (log scale): FR converges faster. (b) Region occupancy under FR relaxes from an unbalanced start toward the balanced 1/3 split (dotted). (c) Cumulative barrier crossings are nearly identical — the benefit here is redistribution, not a large increase in raw crossings. (d) FR score dispersion decays as $\hat{p}_t \rightarrow q_t$.

A notable feature of Figure 5(c) is that FR does *not* produce many more barrier crossings than ABF in this example; the benefit comes from *reallocating* existing replicas toward starved values of x rather than from manufacturing rare transitions. Consistent with this, the event fractions throughout (Figure 2, bottom right) are of order 10^{-5} : in the successful regimes FR is a *weak* correction.

6.2 FR does not visibly corrupt $Y \mid X = x$

The central risk of a configuration-resampling step is that it distorts the conditional law $Y \mid X = x$ that ABF relies on. Figure 6 compares the conditional fidelity of all methods at the final time. The FR variants are *not* worse than ABF only; near the barrier ($x \in \{-0.5, 0, 0.5\}$), where ABF-only under-samples, the FR variants are in fact slightly *better* on both the L^2 and KL conditional metrics. This is direct evidence that, in the successful regimes, the FR mechanism improves x -coverage without corrupting the orthogonal sampling — exactly the benign behaviour Proposition 1 requires.

Remark 2 (Conditional diagnostics are not configuration-resolved). The current conditional-

Conditional sampling fidelity at T (aggregated over configs and seeds; not config-resolved)

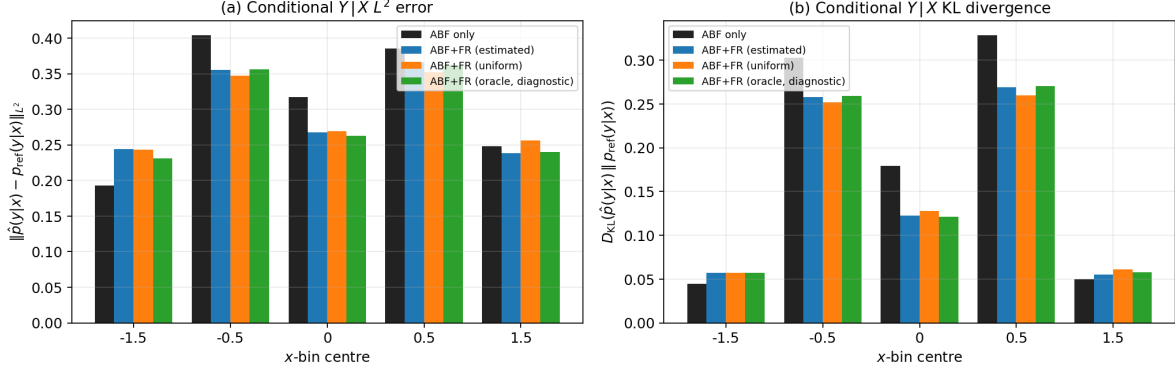


Figure 6: Conditional $Y | X = x$ fidelity at T , by x -bin: (a) L^2 distance and (b) KL divergence of the empirical $\hat{p}(y | x)$ to the reference $p_{\text{ref}}(y | x)$. FR does not inflate the conditional error; near the barrier it slightly reduces it. *Limitation:* the conditional CSV identifies runs only by (method, target, seed) and is aggregated over configurations, so this figure cannot attribute differences to specific good/bad hyperparameters (Section 7).

diagnostics output records only (method, target type, seed) per row, not the FR hyperparameters. It therefore supports the *aggregate* statement that successful FR settings do not visibly corrupt $Y | X = x$, but it cannot isolate a specific “good” versus “bad” configuration’s effect on the conditional. A targeted good-versus-bad attribution would require configuration-resolved conditional output; we list this as a concrete improvement in Section 7.

6.3 What determines a good FR? (Q3)

The results let us name the factors that govern whether FR helps, and how.

Target quality is the practical bottleneck. The estimated target captures 11.5% of the integrated-error reduction against the oracle’s 16.6% (Table 3). Since the estimated and oracle targets share the *same* mechanism and differ only in whether the target free energy is the online estimate or F_{ref} , this gap is attributable to the target estimate, not the birth–death. Improving the online target free-energy estimate (its accuracy and its stability under the EMA) is therefore the most direct route to a better practical FR.

FR strength γ trades speed against final accuracy. Figure 7 (estimated target) shows the integrated error decreasing as γ grows — a stronger correction converges faster — which is why the integrated rule selects the aggressive $\gamma = 0.1$. But the same aggressive setting does *not* minimise the final error (Section 5.4): the final-accuracy rule prefers the gentle $\gamma = 0.01$. There is thus an explicit speed-versus-accuracy trade-off in γ .

Bandwidth η and burn-in b . The marginal bandwidth η sets the resolution at which over/under-population is detected; intermediate values are robust in Figure 7. The burn-in b controls when FR begins: starting FR early (small b) helps the *integrated* error here, because the largest imbalance is at early times, whereas a later burn-in is safer for final accuracy. We caution against universalising

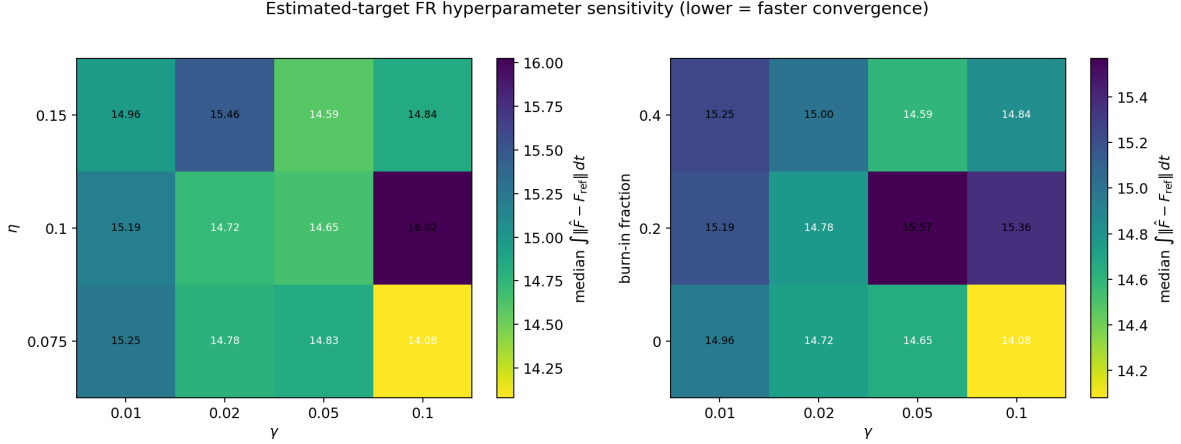


Figure 7: Estimated-target FR hyperparameter sensitivity: median integrated $L^2(F)$ over the grid (lower = faster convergence). Left: γ versus η ; right: γ versus burn-in fraction. The integrated error improves with larger γ and earlier burn-in in this example; intermediate η is robust. These trends inform Q3 but should not be extrapolated beyond this regime.

“start early”: in this example ABF builds a usable target quickly, so an early target is already informative; in harder problems a premature target could misdirect the birth–death.

Event fraction and diversity. Across the successful configurations the event fractions are tiny ($\sim 10^{-5}$) and the cap $f_{\max}$ is never the binding constraint, so the production configurations are *gentle* and particle diversity is preserved (no collapse of ancestors). A more aggressive birth–death — larger γ , larger event fractions — could in principle still harm accuracy by eroding diversity or by acting on a noisy target; the present configurations simply do not reach that regime.

Conditional fidelity. Finally, a good FR must preserve $Y \mid X = x$ (Figure 6). The successful configurations do; this is a necessary condition for the coverage gain to translate into a mean-force gain rather than a bias.

7 Discussion, limitations, and future directions

7.1 Limitations (Q4)

We are deliberately conservative about what this study establishes.

1. **The example is easy.** The model is two-dimensional with a simple linear reaction coordinate $\xi = x$ and a single dominant barrier. The orthogonal coordinate y mixes quickly, so the conditional $Y \mid X = x$ equilibrates fast and the ABF feedback is forgiving. Effects seen here may not transfer to high-dimensional systems or nonlinear reaction coordinates.
2. **ABF only is already a strong baseline.** Because plain ABF already converges well on this potential, the *room* for FR to help is small; this is why the practical improvements are modest (Section 5.7) rather than transformative. A modest improvement over a strong baseline should not be oversold.

3. **The oracle target is not a method.** The oracle uses F_{ref} , the unknown being computed, and is reported only as a diagnostic positive control (Remark 1). Its strong performance must be read as “the FR mechanism has potential when the target direction is accurate”, never as a deployable result. All practical claims rest on the estimated target.
4. **Conditional diagnostics are not configuration-resolved.** As in Remark 2, the conditional $Y \mid X$ output is keyed only by (method, target, seed). It supports the aggregate claim that successful FR does not corrupt the conditional, but cannot causally attribute good-versus-bad behaviour to specific hyperparameters.
5. **The production estimator is a validated surrogate.** The GPU backend uses a binned-smoothed mean-force/marginal estimator and a vectorised fixed- N birth–death. These pass the validation suite and agree with the kernel reference to within tolerance, but they are not mathematically identical to the original kernel notebook implementation (ABF-FR.ipynb); a residual discretisation difference exists by construction.
6. **Harder regimes are untested.** The decisive question — is FR “icing on the cake” (a mild accelerator when ABF already works) or “snow in the winter” (a rescue when ABF alone struggles)? — cannot be answered from an easy example. It requires regimes that stress ABF: larger β (higher barriers), fewer particles, shorter computational budgets, and slower orthogonal mixing (a stiff or metastable $Y \mid X$).

7.2 Failure modes

The analysis points to two concrete ways FR can fail, both consistent with the data. First, a *noisy or biased target*: the estimated–oracle gap (Section 6.3) shows that an inaccurate online target free energy caps the attainable benefit, and a sufficiently wrong target would direct the birth–death toward the wrong regions of x . Second, an *overly aggressive mechanism*: although the present configurations are gentle ($\sim 10^{-5}$ event fractions), large γ or large event fractions could erode particle diversity or perturb $Y \mid X = x$ faster than the dynamics can re-equilibrate it, turning the coverage gain into a conditional bias. The uniform target is a third, milder failure mode of *target misspecification*: it works here only because the converged biased marginal is near-uniform, and would misdirect FR once $B_t \not\approx F$.

7.3 Future directions

- **Stress-test regimes.** Repeat the study at larger β , with fewer particles, shorter budgets, and a stiffer orthogonal coordinate, to determine whether FR is an accelerator or a rescue.
- **Better target estimation.** Since target quality is the practical bottleneck, improve the online estimate of the target free energy — variance reduction, better EMA/smoothing, or uncertainty-aware targets — to close the estimated–oracle gap.
- **Configuration-resolved conditional diagnostics.** Emit conditional $Y \mid X$ statistics per configuration so that good-versus-bad FR can be attributed mechanistically (Remark 2).
- **Adaptive FR strength.** Tie γ and the burn-in to a data-driven signal (e.g. the measured imbalance or target uncertainty) rather than a fixed grid, to obtain the early-time speed of an aggressive rate without sacrificing final accuracy.
- **Higher-dimensional and nonlinear ξ .** Test multi-barrier and multi-dimensional reaction coordinates where the conditional sampling is genuinely hard.

8 Conclusion

On a controlled two-dimensional model with a quadrature-exact reference free energy, a Fisher–Rao birth–death correction can help the Adaptive Biasing Force method, but only as a *carefully controlled marginal reallocation mechanism*. The practical, estimated-target FR improves the time-integrated free-energy error by 11.5% over ABF only at its best configuration and gives smaller final-budget gains; it is best understood here as a convergence accelerator and stabiliser rather than a qualitative rescue. The improvement is not unconditional — it depends on the FR rate, bandwidth and burn-in, and poor choices match or underperform ABF — and across all successful settings FR is *weak* (event fractions of order 10^{-5}), *target-consistent*, and does not visibly corrupt the conditional distribution $Y \mid X = x$ on which ABF depends.

Two interpretive points are essential. First, the practical value of FR is governed by the accuracy and stability of the online target free-energy estimate: the gap between the estimated target (11.5% integrated-error reduction) and the oracle control (16.6%) localises the bottleneck in target estimation, not in the birth–death mechanism. Second, the oracle-target experiments must be read *only* as diagnostic evidence that the mechanism has potential when the target marginal direction is accurate; because the oracle uses the reference free energy F_{ref} , it is not a deployable algorithm and is never recommended as one. Whether FR is “icing on the cake” or “snow in the winter” will be settled not on this easy example but in the harder regimes outlined in Section 7.

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